

Unity requires physically enforced agreement

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Abstract

Human conscious experience is unified: the brain's many local descriptions of a shared situation agree. We argue that unity requires this agreement to be physically enforced, produced and maintained by a coupling through which the graded physical state of each region continuously pulls the regions it overlaps toward agreement. Clocked digital hardware computes agreement instead. Its noise margins make every content readout locally constant in micro state, and we prove that a readout with a margin, under continuous micro dynamics, depends on no region's graded state; content read from bits is therefore never physically enforced, whatever algorithm the bits run. For a coupled sheet of oscillators we show that the width of the coupling kernel sets how fast agreement decays with distance, and that only a kernel tail slower than r^{-4} makes it independent of distance. The premise yields interventional tests on cortex and on artificial systems, and we state the outcomes that would refute it on each.

1 Introduction

Premise P (physical unity). A system’s contents are unified only if the content of each region depends continuously on the graded physical state of the regions it overlaps, through a coupling that contracts their disagreement.

P is a necessary condition for unity. It rejects functionalism about unity: a perfect simulation of a coupled field reproduces the field’s content-level dynamics and fails P. The paper makes three claims:

1. A content readout with a noise margin, under continuous micro dynamics, depends on the graded state of no region, so content read from bits is not physically enforced whatever algorithm the bits run (Section 4).
2. On a noisy coupled sheet, the width of the coupling kernel sets how fast agreement decays with distance, and only a kernel tail slower than r^{-4} makes agreement independent of distance (Section 5).
3. P predicts that a sub-threshold perturbation of one cortical region shifts the content decoded in an overlapping region toward agreement, and that no perturbation below a digital system’s noise margin shifts any of its contents (Section 8).

2 Synchrony is not unity

Equal phases coexist with incompatible local descriptions, and compatible local descriptions determine a unique global state. Unity is therefore a property of what overlapping regions describe, and synchrony is at most a route to it.

3 Unity as agreement over a declared cover

Agreement is measured between regions that decode a shared variable. The cover of regions is fixed by decodability before agreement is measured, so that the measurement cannot be emptied by the choice of cover.

4 Physical enforcement versus routed agreement

A readout has a noise margin on an operating set when every micro state in that set has a neighbourhood on which the readout is constant. Content depends gradedly on a region when no neighbourhood of that region's micro state leaves the next content unchanged. If the micro dynamics is continuous and carries a state into a margin's operating set, no region's micro state enters the next content gradedly (`not_physicallyEnforced_of_margin`). A thresholded continuous quantity has a margin off its threshold, and a finite word of such readouts has one too, so the conclusion holds for any content read from bits (`not_physicallyEnforced_of_bits`).

5 How far a physical coupling enforces agreement

At identical natural frequencies a noisy coupled sheet is a gradient system. In the harmonic approximation the variance of the phase difference between two sites is the noise strength times their effective resistance, with the coupling constants as conductances. Under short-range coupling that resistance grows as the logarithm of distance, with a coefficient inversely proportional to the kernel's second moment; a kernel tail slower than r^{-4} bounds it.

Three facts hold without the harmonic approximation. Adding coupling

never raises the effective resistance (`conductance_mono`), so a long-range tail added to a short-range kernel can only improve agreement, whatever its shape. A direct coupling between two sites caps their resistance at its reciprocal, at any distance (`le_conductance_of_edge`). A Lipschitz encoder carries phase agreement to content agreement: the mean-square content discrepancy is at most the squared Lipschitz constant times the mean-square phase difference (`integral_dist_sq_le`). With the harmonic identity, an encoder of Lipschitz constant L at noise strength D therefore has root-mean-square content discrepancy at most $L\sqrt{D R_{\text{eff}}}$ between two sites of effective resistance R_{eff} .

6 The unity window

Agreement cannot arrive before the causal cone of the coupling allows, and disagreement decays at the relaxation rate of the coupling. Unity at a given time scale requires the interval between the two to be nonempty.

7 Current artificial systems

A transformer running on clocked digital hardware reads every content from bits held within their noise margins, so the result of Section 4 applies to it. Hardware whose contents are graded physical quantities coupled without barriers, such as analog, in-memory or neuromorphic substrates, can satisfy P.

8 Tests

In cortex, weak stimulation of one region below the behavioural threshold should shift the content decoded in an overlapping region, gradedly and toward agreement; an unchanged decoded content in the overlapping region

counts against P. In an artificial system, activation patching below and above the scale at which the logical trajectory changes separates perturbations inside the margin from those that cross it.

9 Limitations

P is a premise and is not derived from the dynamics; the formal results show what follows from it and not that it holds. It is necessary for unity and not sufficient for consciousness: coupled pendulums satisfy it. The margin theorem concerns idealised digital readouts, and graded leakage in real hardware that carries content-relevant agreement would narrow its reach. The identification of phase variance with effective resistance rests on the harmonic approximation at identical natural frequencies, and the growth of resistance with distance is established numerically on finite sheets, not proved. Human data can falsify P but cannot by themselves confirm it against functionalism, because graded coupling and information flow co-vary in every manipulation available so far.